

PDMWorks API (.NET, legacy) — Examples (1/1): Create_Search_Criteria_and_Search_Vault_- Welcome-pdmworksapi

SOLIDWORKS API Documentation (converted from CHM)

Create_Search_Criteria_and_Search_Vault_Example_VB (Topic)

SOLIDWORKS Workgroup PDM API Help

Create Search Criteria and Search Vault Example (VBA)

This example shows how to create a search criteria and search the SOLIDWORKS Workgroup PDM vault using that criteria.

```
Private Sub Command1_Click()  
Dim Options As PDMWorks.PDMWSearchOptions  
Dim Criteria As PDMWorks.PDMWSearchCriteria  
Dim Results As PDMWorks.PDMWSearchResults  
Dim Result As PDMWorks.PDMWSearchResult  
Dim i As Integer  
Dim cnt As Integer  
On Error Resume Next  
  
Set Options = Connection.GetSearchOptionsObject  
If Not Options Is Nothing Then  
Options.IgnoreCase = True  
Options.IgnoreLinks = False  
Options.IncludeHiddenDocuments = True  
Options.SearchConfigSpecificProperties = False  
Options.SearchOnlyChildrenOf = ""  
Options.SearchCriteria.AddCriteria pdmwOr, pdmwDocumentName, , Contains, "sldprt"  
Options.SearchCriteria.AddCriteria pdmwAnd, PDMWConfiguration, "", Contains, "default"  
  
Options.SearchCriteria.SaveToFile "c:\temp\cri_test.sqy"  
Options.SearchCriteria.LoadFromFile "c:\temp\cri_test.sqy"  
  
Set Results = Connection.Search(Options)  
If Not Results Is Nothing Then  
cnt = Results.Count  
  
For i = 0 To cnt - 1  
Set Result = Results(i)  
AddItemToGrid Result  
Next i  
End If  
End If  
  
End Sub
```

Extract Embedded eDrawings Data Example_VB (Topic)

SOLIDWORKS Workgroup PDM API Help

Extract Embedded eDrawings Data Example (VBA)

This example shows how to search drawings in the SOLIDWORKS Workgroup PDM vault, check to see if the drawings of a certain lifecycle status (APPROVED), and if both criteria are met, extract the embedded eDrawings data from the drawings and save it to a local disk.

```
Sub main()  
    On Error GoTo errorHandler  
    Dim connection As PDMWConnection  
    Set connection = CreateObject("PDMWorks.PDMWConnection")  
    connection.Login "pdmwadmin", "pdmwadmin", "localhost"  
    Dim savePath As String  
    savePath = "C:\edrawings\" "Must be a valid path"  
    Dim alldocs As PDMWDocuments  
    Set alldocs = connection.Documents  
    Dim adoc As PDMWDocument  
    For Each adoc In alldocs  
        If LCase(Right(adoc.Name, 6)) = "slddrw" Then  
            If adoc.GetStatus = "APPROVED" Then  
                adoc.SaveEmbeddedEDW savePath  
            End If  
        End If  
    Next  
    connection.Logout  
Exit Sub  
errorHandler:  
    Debug.Print "Error: " & adoc.Name & " : " & Err.Description  
    Resume Next  
End Sub
```

Get_Names_Projects_for_Documents_in_Vault_Example_VBNET (Topic)

SOLIDWORKS Workgroup PDM API Help

Get Names of Projects for All Documents in Vault (VB.NET)

This example shows how to get the names of the projects for all of the documents in the vault. The total number of documents and the names of their projects are written to the console window.

Imports PDMWorks

Module Module1

Sub Main()

Dim PDMW_conn As PDMWorks.PDMWConnection

PDMW_conn = CreateObject("PDMWorks.PDMWConnection")

PDMW_conn.**Login**("pdmwadmin", "pdmwadmin", "localhost")

Dim allDocs As PDMWDocuments

allDocs = PDMW_conn.**Documents**

Console.WriteLine("Total Documents: " + allDocs.**Count**.ToString)

' Given a PDMWDocuments collection called doc

For Each doc As PDMWDocument In allDocs

Dim Project As String

Project = doc.**project**

Console.WriteLine("Document Project: " + Project)

Next

PDMW_conn.**Logout**()

End Sub

End Module

Get Size of Document Larger Than 2GB Example VB (Topic)

SOLIDWORKS Workgroup PDM API Help

Get Size of Document Larger Than 2GB Example (VBA)

You can get the size of a document larger than 2GB by calling either:

- IPDMWDocument::GetDocumentSize
- or -
- IPDMWDocument::SizeH and IPDMWDocument::Size and combining their return values.

This example shows how to get the size of a document larger than 2GB, using both ways.

```
' -----  
' Preconditions:  
' 1. Replace <your_2GB+_document> with the name of a document that is larger  
'    than 2GB and that is checked into your vault.  
' 2. Open the Immediate window.  
'  
' Postconditions: Examine the Immediate window.  
' -----
```

Option Explicit

Sub main()

```
    Dim connection As PDMWConnection  
    Dim doc As PDMWDocument  
    Dim fileName As String  
  
    Dim lSizeHi As Long  
    Dim lSizeLow As Long  
    Dim docSize As Double  
  
    ' Log into vault  
    Set connection = CreateObject("PDMWorks.PDMWConnection")  
    connection.Login "pdmwadmin", "pdmwadmin", "localhost"  
  
    fileName = "<your_2GB+_document>"  
  
    ' Name of 2GB+ document in vault  
    Set doc = connection.GetSpecificDocument(fileName)  
  
    ' Using new IPDMWDocument::GetDocumentSize method  
    docSize = CDBl(doc.GetDocumentSize)  
    Debug.Print ("Result using IPDMWDocument::GetDocumentSize method:")  
    Debug.Print ("  Size of " + fileName + " is " + Str(docSize) + " bytes")  
    Debug.Print  
  
    ' Using both IPDMWDocument::Size and IPDMWDocument::SizeH properties  
    '  
    ' NOTE: Because VB's long is signed, you must convert the long to  
    ' a positive value for both return values  
    lSizeHi = doc.SizeH  
    docSize = IIf(lSizeHi < 0, 4294967296# + lSizeHi, lSizeHi)  
    docSize = docSize * 4294967296#  
  
    lSizeLow = doc.Size  
    docSize = docSize + IIf(lSizeLow < 0, 4294967296# + lSizeLow, lSizeLow)  
  
    Debug.Print ("Result using IPDMWDocument::Size and IPDMWDocument::SizeH properties:")  
    Debug.Print ("  Size of " + fileName + " is " + Str(docSize) + " bytes")  
    Debug.Print
```

connection.Logout

End Sub

Get_Specific_Document_Example_VB (Topic)

SOLIDWORKS Workgroup PDM API Help

Get Specific Document Example (VBA)

This example shows how to log into a SOLIDWORKS Workgroup PDM vault, get a collection of documents, and get a specific document.

```
Sub main()  
    Dim connection As PDMWConnection  
    Set connection = CreateObject("PDMWorks.PDMWConnection")  
    connection.Login "pdmwadmin", "pdmwadmin", "localhost"  
  
    Dim alldocs As PDMWDocuments  
    Dim doc1 As PDMWDocument  
    Dim doc2 As PDMWDocument  
    Dim doc3 As PDMWDocument  
    Set alldocs = connection.Documents  
    Set doc1 = connection.Documents.Item(0)  
    Set doc2 = connection.GetSpecificDocument("bracket.sldprt")  
    "Look at alldocs, doc1, and doc2 in the debugger"  
    Set doc3 = alldocs("bracket.sldprt")  
  
    connection.Logout  
End Sub
```

GettingStarted-pdmworksapi (Topic)

SOLIDWORKS Workgroup PDM API Help

Getting Started

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This topic contains the following information:

- [Application Basics](#)
- [Error Handling](#)
- [Obsolete APIs](#)

Application Basics

Writing a SOLIDWORKS Workgroup PDM API application typically involves:

1. Instantiating a SOLIDWORKS Workgroup PDM connection and logging in.

When you connect and log in to SOLIDWORKS Workgroup PDM via the API, the SOLIDWORKS Workgroup PDM API object model is implemented as a snapshot of what is in the vault at that time. If the state of the vault changes either through the user interface or the API (for example, by a check in, the creation of a new project, the addition of a note, and so on), you must call [IPDMWConnection::Refresh](#) to ensure that subsequent API calls return the most current data.

IMPORTANT: Because [IPDMWConnection::CheckIn](#) is dependent on **pdmworks_libFNP.dll**, you must ensure that your application can find this DLL. The simplest way for your application to find **pdmworks_libFNP.dll** is to add its location to the PATH environmental variable. See [Microsoft Corp's Dynamic-Link Library Search Order](#) for details.

SOLIDWORKS Workgroup PDM API contains several methods that change the state of the SOLIDWORKS Workgroup PDM vault; however, most of the exposed object model contains read-only properties. Therefore, a typical SOLIDWORKS Workgroup PDM application displays and analyzes vault data rather than updates it.

2. Getting and reading the various VB-compatible collections that represent:
 - projects,
 - documents in those projects, and
 - those projects' configurations, references, and other properties.
3. Logging out.

The SOLIDWORKS Workgroup PDM API exposes VB-compatible collections for easy iteration of similar objects using the **For Each...Next** statement. For example, to iterate over the set of documents in a vault, you could use the following code fragment:

```
Dim connection As PDMWConnection
Dim allDocs As PDMWDocuments
Dim document As PDMWDocument

Sub main()
    Set connection = CreateObject("PDMWorks.PDMWConnection")
    connection.Login "pdmwadmin", "pdmwadmin", "vault_server_name"
    Set allDocs = connection.documents

    For Each document In allDocs
        'Access each PDMWDocument here
    Next
    connection.Logout

End Sub
```

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Error Handling

When appropriate, the SOLIDWORKS Workgroup PDM API returns an error status code, for example, when performing [IPDMWProperties::Update](#). Sometimes, directly returning a status code is not possible, for example, when retrieving a document via [IPDMWConnection::GetSpecificDocument](#).

If an error status is returned, it is in the form of a SOLIDWORKS Workgroup PDM error code (a 32-bit value). This code returns 0 for success and non-0 for error.

SOLIDWORKS recommends that you use the intrinsic Visual Basic Err object to get rich error information. In this case, the Err.number field does not correspond directly to the SOLIDWORKS Workgroup PDM error code, but mappings back to the error code are provided. The Err.Description field is filled with a descriptive text field indicating the error. In this way, better tracing of errors can be done through the Err object.

In Visual Basic 5.0, error handling can easily be accomplished with the **On Error Resume Next** statement. For example:

```
On Error Resume Next
```

```
Dim conn
```

```
Set conn = CreateObject(PDMWorks.PDMWConnection)
```

```
conn.Login user, incorrect password, vault
```

```
Err.number contains the error number
```

```
Err.description contains the appropriate error description
```

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Obsolete APIs

SOLIDWORKS Workgroup PDM is continually enhancing its APIs. In most instances, when an existing API is superseded by a new API, the new API has the same name as the existing API followed by a number. A higher number indicates a more recent API.

Superseded APIs are not supported. Help topics for superseded APIs are marked as obsolete.

SOLIDWORKS recommends that you use the newest APIs in your code, and, whenever possible, update your existing code to use the newest APIs.

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List Assemblies Where Part Used Example_VB (Topic)

SOLIDWORKS Workgroup PDM API Help

List Assemblies Where Part Used Example (VBA)

This example shows how to look at each part in the SOLIDWORKS Workgroup PDM vault and list the assemblies in which the part is used.

```
Sub main()
    Dim connection As PDMWConnection
    Set connection = CreateObject("PDMWorks.PDMWConnection")
    connection.Login "pdmwadmin", "pdmwadmin", "localhost"
    Dim alldocs As PDMWDDocuments
    Set alldocs = connection.Documents
    Dim adoc As PDMWDocument
    Dim msgBoxStr As String
    For Each adoc In alldocs
        If LCase(Right(adoc.Name, 6)) = "sldprt" Then
            msgBoxStr = msgBoxStr & "Part: " & adoc.Name & vbNewLine
            msgBoxStr = msgBoxStr & "Assemblies where it is used:" & vbNewLine
            Dim whereUsedList As PDMWLinks
            Set whereUsedList = adoc.WhereUsed
            Dim I As PDMWLink
            For Each I In whereUsedList
                If LCase(Right(I.Document.Name, 6)) = "sldasm" Then
                    msgBoxStr = msgBoxStr & "    " & I.Document.Name & vbNewLine
                ' Stop
            End If
        Next
        msgBoxStr = msgBoxStr & "----" & vbNewLine
    End If
Next
Output.Data = msgBoxStr
Output.Show
connection.Logout
End Sub
```

PDMWorks.Interop.pdmworks (Topic)

PDMWorks.Interop.pdmworks Assembly



Namespaces

Namespace	Description
PDMWorks.Interop.pdmworks	SOLIDWORKS Workgroup PDM API

ReleaseNotes-pdmworksapi (Topic)

SOLIDWORKS Workgroup PDM API Help

Release Notes

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No new functionality was introduced in this version of SOLIDWORKS Workgroup PDM API.

Welcome-pdmworksapi (Topic)

SOLIDWORKS Workgroup PDM API Help

Welcome

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SOLIDWORKS Workgroup PDM is a project-data management application designed to run both natively inside of the SOLIDWORKS environment and as a standalone application.

The SOLIDWORKS Workgroup PDM Application Programming Interface (API) is an OLE programming interface to SOLIDWORKS Workgroup PDM, which allows you to automate and customize SOLIDWORKS Workgroup PDM. The API contains functions that you can call from Visual Basic for Applications (VBA), VB.NET, C++ 6.0, C++/CLI, and C#. These functions provide direct access to the SOLIDWORKS Workgroup PDM environment. It is assumed that you are familiar with at least one of these programming languages.

This online reference manual documents the objects and collections available in the SOLIDWORKS Workgroup PDM API, including their associated methods and properties.

You need WinZip installed on your computer to download the sample project. See <http://www.winzip.com> for details.

- All of the sample code supplied is provided on an as-is basis, and is only intended to demonstrate ways of using the SOLIDWORKS Workgroup PDM API. SOLIDWORKS makes no representations or warranties regarding these samples. Any licensed user of SOLIDWORKS Workgroup PDM is free to use any or all of these samples in connection with building applications related to SOLIDWORKS Workgroup PDM, and is granted a royalty free, non-exclusive license for these samples, or parts thereof. Intellectual property rights of the samples remain with SOLIDWORKS. Any confidentiality provisions of the SOLIDWORKS Workgroup PDM license apply to the samples.
- Some sample code might contain obsolete APIs.